

CompTIA A+ Core 1 Playbook

IT Labs

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4.2.6 Troubleshoot Boot Issues

You work at a computer repair store. A customer has brought in a computer for repairs. It has the following specifications:

- Intel i7-960 processor
- 16 GB of DDR4 memory
- Three SATA hard drives (one with the operating system installed and the others for storing data)
- A single SATA optical (DVD) drive

You replaced the power supply in the computer, and now the computer won't boot.

In this lab, your task is to diagnose and correct the problems with this system.

You'll know that you've successfully corrected all the problems when the computer successfully boots to Windows.

In this scenario, all the problems are things that could occur when replacing a power supply.

Complete this lab as follows:

1. Replicate the problem by turning the computer on.
 1. On the computer, select the **power button** to turn the computer on.
A prompt is shown, indicating that the computer failed to turn on.
 2. Select **OK** to acknowledge the prompt.
2. Provide power to the computer.
 1. Switch to the **Back** view of the computer.
 2. Select the **AC Power Connector** plugged into the computer.
 3. Verify that the other end is plugged into the wall plate.
 4. Select the **power switch** on the power supply to provide power to the computer.
 5. Switch to the **Front** view of the computer.
 6. Select the **power button** on the computer to turn it on.
A prompt is shown, indicating that the computer failed to turn on.
 7. Select **OK** to acknowledge the prompt.
3. Check for power to the motherboard and disks.
 1. Switch to the **Motherboard** view of the computer.
 2. Select the **power supply**.
Notice that the CPU has no power connector attached. Also, notice that the primary disk has no power attached either.
 3. From the Selected Component pane:
 - Drag the **Connector, Power Supply, 8-pin CPU** connector to the power port (upper left) on the motherboard.
 - Drag an unconnected **Connector, SATA, 15-pin** cable to the top disk (the left port).
4. Provide power to the computer and turn it on.
 1. Switch to the **Front** view of the computer.
 2. Select the **power button** on the computer to turn it on.
The computer powers on, but no bootable device is found.
5. Check for proper SATA cable connections.
 1. From the upper left, select **Workbench**.
 2. Select the **power button** to turn the computer off.
 3. Switch to the **Motherboard** view of the computer.
 4. For the top disk, select the **SATA cable**.
Notice that the other end of the cable is disconnected.
 5. From the Selected Component pane, drag the **Connector, SATA, 7-pin** connector to a SATA port on the motherboard.
6. Test the solution.
 1. Switch to the **Front** view of the computer.
 2. Select the **power button** on the computer to turn it on.
The computer boots to Windows.